Normal appearing white matter (NAWM) - Insight46 phase 1

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Document details

Categories of variables	Word in noise
Summary of work undertaken	- Individual .csv Word in noise files amalgamated into one spreadsheet - Participants who did not undergo testing values coded as -99 (in each case this was due to functioning testing equipment not being available on the day) - All individual .csv files were checked all fields were completed and labelled with correct participant number by TP in an ongoing manner throughout testing - Please see methods section below for more details of testing
Names of source variables	visit_date, of, visit), sTime_norm1, time, of, test, normal, condition), finish_norm1, time, of, test, normal, condition), list_norm1, list, used, for, test, normal, condition), mean_norm1, SRT, for, test, normal, condition), sd_norm1, deviation, for, test, normal, condition), reversals_norm1, of, reversals, for, test, normal, condition), trials_norm1, of, trials, for, test, normal, condition), sTime_norm2, time, of, test, normal, condition), finish_norm2, time, of, test, normal, condition), list_norm2, list, used, for, test, normal, condition), mean_norm2, SRT, for, test, normal, condition), sd_norm2, deviation, for, test, normal, condition), reversals_norm2, of, reversals, for, test, normal, condition), trials_norm2, of, trials, for, test, normal, condition), sTime_nal1, time, of, test, hearing, aid, formula, condition), finish_nal1, time, of, test, hearing, aid, formula, condition), list_nal1, list, used, for, test, hearing, aid, formula, condition), mean_nal1, SRT, for, test, hearing, aid, formula, condition), sd_nal1, deviation, for, test, hearing, aid, formula, condition), reversals_nal1, of, reversals, for, test, hearing, aid, formula, condition), trials_nal1, of, trials, for, test, hearing, aid, formula, condition), sTime_nal2, time, of, test, hearing, aid, formula, condition), finish_nal2, time, of, test, hearing, aid, formula, condition), list_nal2, list, used, for, test, hearing, aid, formula, condition), mean_nal2, SRT, for, test, hearing, aid, formula, condition), sd_nal2, deviation, for, test, hearing, aid, formula, condition), reversals_nal2, of, reversals, for, test, hearing, aid, formula, condition), trials_nal2, of, trials, for, test, hearing, aid, formula, condition), nal_complete, NAL, was, completed, or, not)
Output variables	visit_date_i46p1, stime_norm1_i46p1, finish_norm1_i46p1, list_norm1_i46p1, mean_norm1_i46p1, sd_norm1_i46p1, reversals_norm1_i46p1, trials_norm1_i46p1, stime_norm2_i46p1, list_norm2_i46p1, finish_norm2_i46p1, mean_norm2_i46p1, sd_norm2_i46p1, reversals_norm2_i46p1, trials_norm2_i46p1, stime_nal1_i46p1, list_nal1_i46p1, finish_nal1_i46p1, mean_nal1_i46p1, sd_nal1_i46p1, reversals_nal1_i46p1, trials_nal1_i46p1, stime_nal2_i46p1, list_nal2_i46p1, finish_nal2_i46p1, mean_nal2_i46p1, sd_nal2_i46p1, reversals_nal2_i46p1, trials_nal2_i46p1, nal_complete_i46p1

Methods

357 out of the 368 participants analysed underwent two trials of a word-in-multi-talker babble identification paradigm whereby high frequency monosyllabic words were presented in a random order with a multi-talker babble (composed of 20 voices presented at a fixed level of 65 dB SPL) in the background The sound level of each individual word varied with each presentation as defined by a signal to noise ratio (SNR). After each word the participant was asked to repeat verbatim what they heard. The experimenter then recorded the response as correct or incorrect. Each trial commenced with an SNR of 12 dB and tracked 50% correct adaptively with a one-up one-down rule (maximum SNR 18 dB). The test ended after 8 reversals or 25 trials, with the Speech Reception Threshold (SRT: the SNR that leads to approximately 50% correct answers) calculated from the mean of the reversals. To rule out possible contributions of differences in audiometric thresholds above 4 kHz, the stimuli were low-pass filtered at 4 kHz using a 4th order Butterworth filter. Eight different lists of 25 highly frequency monosyllabic words were available and were chosen at random for each trial. No one list was used more than once for each participant. In 129 out of 140 participants with evidence of hearing impairment (PTA best hearing ear > 25 dB HL) two additional trials were performed where the stimuli were spectrally shaped using the National Acoustics Laboratories-Revised (NAL-R) linear prescriptive formula based on their individual

thresholds (Byrne and Dillon, 1986; Schoof and Rosen, 2014). The order in which these trials were performed relative to the non-NAL-R assisted trials was random for each subject to exclude the possibility of learning effects influencing the results. The participants were not informed of the change in nature of the auditory stimuli if a NAL-R formula had been applied. The participants were seated in a quiet room and stimuli were presented binaurally over ATH-M50 professional studio monitor headphones via MATLAB software on a Lenovo Yoga Ultrabook with Intel inside CORE i7 processer with Focusrite Scarlett 2i2 external USB soundcard. Participants were instructed to keep their eyes open throughout the testing.

Raw variables

Please note final variable names can be found at the bottom of the page. Raw variables list with variable labels:

- visit_date (date of visit)
- sTime_norm1 (start time of test 1 – normal condition)
- finish_norm1 (finish time of test 1 – normal condition)
- list_norm1 (word list used for test 1 – normal condition)
- mean_norm1 (mean SRT for test 1 – normal condition)
- sd_norm1 (standard deviation for test 1 – normal condition)
- reversals_norm1 (number of reversals for test 1 – normal condition)
- trials_norm1 (number of trials for test 1 – normal condition)
- sTime_norm2 (start time of test 2 – normal condition)
- finish_norm2 (finish time of test 2 – normal condition)
- list_norm2 (word list used for test 2 – normal condition)
- mean_norm2 (mean SRT for test 2 – normal condition)
- sd_norm2 (standard deviation for test 2 – normal condition)
- reversals_norm2 (number of reversals for test 2 – normal condition)
- trials_norm2 (number of trials for test 2 – normal condition)
- sTime_nal1 (start time of test 1 – hearing aid formula condition)
- finish_nal1 (finish time of test 1 – hearing aid formula condition)
- list_nal1 (word list used for test 1 – hearing aid formula condition)
- mean_nal1 (mean SRT for test 1 – hearing aid formula condition)
- sd_nal1 (standard deviation for test 1 – hearing aid formula condition)
- reversals_nal1 (number of reversals for test 1 – hearing aid formula condition)
- trials_nal1 (number of trials for test 1 – hearing aid formula condition)
- sTime_nal2 (start time of test 2 – hearing aid formula condition)
- finish_nal2 (finish time of test 2 – hearing aid formula condition)
- list_nal2 (word list used for test 2 – hearing aid formula condition)
- mean_nal2 (mean SRT for test 2 – hearing aid formula condition)
- sd_nal2 (standard deviation for test 2 – hearing aid formula condition)
- reversals_nal2 (number of reversals for test 2 – hearing aid formula condition)
- trials_nal2 (number of trials for test 2 – hearing aid formula condition)
- nal_complete (whether NAL was completed or not)

Variables in this document

29 high-confidence matches

Variable	Label	How it was found
Sub-Studies [57] › Insight46 [699] › Physical measures [6035] › Hearing test [60354]		

Variable	Label	How it was found
finish_nal1_i46p1	Finish time of test 1 - hearing aid formula condition - word in noise - Insight46 phase 1	topsheet field
finish_nal2_i46p1	Finish time of test 2 - hearing aid formula condition - word in noise - Insight46 phase 1	topsheet field
finish_norm1_i46p1	Finish time of test 1 - normal condition - word in noise - Insight46 phase 1	topsheet field
finish_norm2_i46p1	Finish time of test 2 - normal condition - word in noise - Insight46 phase 1	topsheet field
list_nal1_i46p1	Word list used for test 1 - hearing aid formula condition - word in noise - Insight46 phase 1	topsheet field
list_nal2_i46p1	Word list used for test 2 - hearing aid formula condition - word in noise - Insight46 phase 1	topsheet field
list_norm1_i46p1	Word list used for test 1 - normal condition - word in noise - Insight46 phase 1	topsheet field
list_norm2_i46p1	Word list used for test 2 - normal condition - word in noise - Insight46 phase 1	topsheet field
mean_nal1_i46p1	Mean speech reception threshold (SRT) for test 1 - hearing aid formula condition - word in noise - Insight46 phase 1	topsheet field
mean_nal2_i46p1	Mean speech reception threshold (SRT) for test 2 - hearing aid formula condition - word in noise - Insight46 phase 1	topsheet field
mean_norm1_i46p1	Mean speech reception threshold (SRT) for test 1 - normal condition - word in noise - Insight46 phase 1	topsheet field
mean_norm2_i46p1	Mean speech reception threshold (SRT) for test 2 - normal condition - word in noise - Insight46 phase 1	topsheet field
nal_complete_i46p1	Whether National Acoustics Laboratories-Revised (NAL-R) linear prescriptive formula (NAL) was completed or not - word in noise - Insight46 phase 1	topsheet field
reversals_nal1_i46p1	Number of reversals for test 1 - hearing aid formula condition - word in noise - Insight46 phase 1	topsheet field
reversals_nal2_i46p1	Number of reversals for test 2 - hearing aid formula condition - word in noise - Insight46 phase 1	topsheet field
reversals_norm1_i46p1	Number of reversals for test 1 - normal condition - word in noise - Insight46 phase 1	topsheet field
reversals_norm2_i46p1	Number of reversals for test 2 - normal condition - word in noise - Insight46 phase 1	topsheet field
sd_nal1_i46p1	Standard deviation for test 1 - hearing aid formula condition - word in noise - Insight46 phase 1	topsheet field
sd_nal2_i46p1	Standard deviation for test 2 - hearing aid formula condition - word in noise - Insight46 phase 1	topsheet field
sd_norm1_i46p1	Standard deviation for test 1 - normal condition - word in noise - Insight46 phase 1	topsheet field
sd_norm2_i46p1	Standard deviation for test 2 - normal condition - word in noise - Insight46 phase 1	topsheet field
stime_nal1_i46p1	Start time of test 1 - hearing aid formula condition - word in noise - Insight46 phase 1	topsheet field
stime_nal2_i46p1	Start time of test 2 - hearing aid formula condition - word in noise - Insight46 phase 1	topsheet field
stime_norm1_i46p1	Start time of test 1 - normal condition - word in noise - Insight46 phase 1	topsheet field

Variable	Label	How it was found
stime_norm2_i46p1	Start time of test 2 - normal condition - word in noise - Insight46 phase 1	topsheet field
trials_nal1_i46p1	Number of trials for test 1 - hearing aid formula condition - word in noise - Insight46 phase 1	topsheet field
trials_nal2_i46p1	Number of trials for test 2 - hearing aid formula condition - word in noise - Insight46 phase 1	topsheet field
trials_norm1_i46p1	Number of trials for test 1 - normal condition - word in noise - Insight46 phase 1	topsheet field
trials_norm2_i46p1	Number of trials for test 2 - normal condition - word in noise - Insight46 phase 1	topsheet field

1 medium-confidence match

Variable	Label	How it was found
Questionnaire data [55] › Health and medical history [510] › Medical conditions [5101] › Heart and circulatory conditions [51014]		
speech	In the last ten years that is since you were 43 years old, have you had a sudden speech problem which got better after a day?- at age 53 years	prose scan

1 low-confidence match

Variable	Label	How it was found
Questionnaire data [55] › Mental health [520] › Present State Examination [5205]		
snr	Subscore of specific neurotic syndromes (for 40 items score is derived from syndromes SD, GA, SA and ED) - at age 36 years	prose scan